

Deep Learning-based Plant Disease Detection and Severity Estimation: A Comprehensive Framework for Precision Agriculture

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Abstract— With the rapid spread of plant diseases, the stability of the food production process worldwide is under more and more threat, and advanced and high-throughput diagnostic tools are needed. Conventional phytopathology is based on manual and expert-visual inspection, which is also subjective and time consuming and geographically limited. In this paper, a new and more detailed machine learning system that automatizes the procedure of detecting and quantifying plant diseases severity is presented. The system uses the Transfer Learning and fine-tuned type of ResNet50 Convolutional Neural Network (CNN) because it was trained on the PlantVillage data only, and it provides high accuracy in the classification of 49 separate plant-disease categories. The major innovation that we have is the Automated Severity Estimation module which makes use of HSV color space analysis and digital image processing to mathematically measure the percentage of infected leaf area to give an actionable but non-subjective measure of disease progression. Moreover, the system includes a post-classification consultative module that is based on Generative AI to provide systematic organic treatment plans and prophylactic actions. The competitive validation accuracy of the proposed framework was 81.07%, which proves that it can be considered a scalable and reliable tool to assist with data-driven decision-making in precision agriculture.

Keywords— Deep Learning, Convolutional Neural Networks, ResNet50, Plant Disease Detection, Severity Estimation, Precision Agriculture, Image Processing, Transfer Learning.

I. INTRODUCTION

Most developing countries depend on agriculture as an economic activity, and it is the main source of livelihood to a large percentage of the world population. Nevertheless, the biotic stresses such as fungal, bacterial, and viral infections always pose a threat to the stability of this sector. The Food and Agriculture Organization (FAO) estimates that crop production in the world suffers a 20-40% annuities due to plant pests and diseases. This does not only affect the economic stability of the farmers; it also poses a problem to the global food security [1].

The most crucial procedure in dealing with the threats is timely and accurate diagnosis. In the past, it has been a manual process, which involves the naked eye and experience of farmers or the agricultural extension officers [2]. Although it works in the small-scale environment, this method has critical drawbacks. It is not objective but by its very nature; various professionals can have different interpretations of the

symptoms [3]. It is also time consuming and intensive and therefore it is not possible to monitor big fields. In addition, access to expert advice is very low in most rural areas, which exposes the farmers to sudden cases of disease outbreaks.

Precision Agriculture has been given a new direction with the introduction of Artificial Intelligence (AI), which is the Computer Vision (CV). Convolutional Neural Networks (CNNs) have shown to be highly successful at learning intricate hierarchical features of image data (learns better than people in many visual recognition problems) [4]. Nevertheless, the majority of the automated systems that are in place today are a binary classifier that defines a plant to be either healthy or diseased without any additional information [5].

The solution in this research will fulfill three essential requirements that are end-to-end:

1. **Specific Disease Detection:** A refined ResNet50 architecture was used to discover particular diseases in a variety of crops (Apple, Corn, Grape, Potato, Tomato, etc.).
2. **Measurement of the extent of the affected leaf area quantitatively:** The application of digital image processing algorithm to estimate the recent percentage of the leaf area. This is more than qualitative terms (such as Bad) to quantitative measurements (such as 67.4% Infected), where it is possible to be more specific in intervention.
3. **Actionable Advice:** Incorporating a Generative AI application, which serves as an artificial agronomist, offering dozens of organic cures, preventive aid, and products recommendations basing solely on the diagnosis of the CNN model.

The synthesis of all these elements into one cohesive web based system will democratize the use of expert level plant pathology applications that will enable farmers to make evidence-based decisions that will result in increased productivity and sustainability [6].

The additional structure of the article was developed with the thorough literature background in Section 2. The rationale of the choice of the system tool and the challenges are then followed in Section 3. In Section 4, the system architecture and methodology is examined. Section 5 is the arrangement of the

obtained results and discussions. Parametric results are obtained and these are concluded with at the end of the journal.

II. LITERATURE REVIEW

Chen et al. (2020) introduced a preliminary deep learning system on leaf disease detection based on an adjusted InceptionV3 framework trained on the PlantVillage dataset. They showed good performance when light was controlled but failed when they were presented with heterogeneous images in the real world. According to the authors, deep models have been observed to be unable to generalize between multiple environments despite being trained on a single-source dataset, and that dataset expansion and augmentation strategies should be implemented to enhance their robustness [7]. Several CNN models, such as VGG, AlexNet, and ResNet, were explored by Ferantinos (2018) to classify multi-crop diseases. The variants of ResNet were found to be more accurate in most of the evaluated models because they were able to learn fine-grained lesion patterns by using residual skip connections. Nonetheless, the research has underscored the fact that CNNs are most effective when they are trained with multiple sets of data because PlantVillage alone is deprived of a field-level variation of shadows, soil backgrounds, and intermittent lighting [8].

Too et al. (2019) have made a wide-scale comparative study of the transfer learning framework (ResNet50, DenseNet121, InceptionResNetV2) to identify the plant diseases. They showed in their experiments that ResNet50 is best suited in accuracy and computational efficiency, especially where the disease category has subtle color-texture detail. One of the main limitations that was observed is that the models were sensitive to domain shift, which implies that they need more large datasets to be trained rather than to become overtrained to controlled lab-style leaf images [9].

Arsenovic et al. (2021) presented the PlantDisease-Real data and suggested the usage of Generative Adversarial Networks (GANs) to produce the artificial images of the field. Their contribution bridged the gap in their field with the controlled condition by ensuring an increase in the dataset with real-world noises, background, and lighting variations. Despite the fact that GAN-based augmentation enhanced the generalization, issues related to the complexity of training and instability were also serious disadvantages in the large-scale agricultural applications [10].

Zhang et al. (2022) suggested a dual-branch CNN which is able to detect disease and provide the localization of lesions. They used attention to emphasize the infected areas with their method, enhancing interpretation and the estimation of severity. Nonetheless, the authors have found that high-quality pixel-wise annotations cannot be easily and cost-effectively acquired to accomplish high-quality lesion segmentation [11]. Kumar et al. (2023) explored the role of dataset fusion, that is, a combination of PlantVillage with a second dataset in the form of a real-world dataset. Their experiments showed that the multi-source training is very effective in increasing the robustness to noise and variability of leaf but also decreases the accuracy that is to be achieved because of the intra-class diversity. This is consistent with what was done in our system

whereby PlantVillage and DrPlantify were combined to give a more enriched dataset of 49 classes [12].

Li et al. (2024) introduced a ResNet based classifier to be used in mobile phones to work in precision agriculture. Their framework delivered high-performance inference with efficiency on edge hardware and at the same time, their system had competitive accuracy. The authors pointed out that the severity estimation is not studied well in existing literature and suggested to use traditional image processing methods with deep learning classifiers as scalable solutions, which is consistent with the severity module of our proposed framework [13].

S. M. Hassan *et al* presented a lightweight ResNet-based classifier optimized for mobile devices aimed at precision agriculture. Their system achieved efficient inference times on edge hardware while maintaining competitive accuracy. The authors noted that severity estimation remains underexplored in current literature and recommended combining traditional image processing techniques with deep learning classifiers for scalable solutions, an approach aligned with the severity module in our proposed framework [14].

III. SYSTEM DESIGN

The suggested system adheres to a modular structure that divides the data capture process, deep learning inference, severity quantification and advisory generation into clear functional units. The design is geared towards scalability, maintenance and capability of adding new features without interfering with the current parts.

A. System Overview

The system is designed to have four significant functional blocks:

1. **Data Pre-processing and Acquisition Module.** This module manages the user uploaded leaf images and carries out the necessary preprocessing functions which include; resizing, normalization, noise filtering and quality verification. As the model will be trained on a fused dataset that will be formed by the concatenation of PlantVillage and DrPlantify, the pre-processing pipeline will guarantee consistency between training and inference inputs.
2. **AI Engine Module**
The fundamental computational unit and it has three internal processes:
 - Image Pre-Processing: converts the input to a normalized tensor which is another form of the input that the model inferences.
 - Feature Learning and Classification: makes use of a refined model version of ResNet50 to categorize the input into 49 plant-disease types constructed out of the merged datasets.
 - Severity Estimation: uses a segmentation algorithm based on HSV and isolates the infected areas and calculates the percentage of leaf area cover.

3. **Database and History Management Module**
This module holds user entry, estimate output, rate of severity and time stamp. It also allows access to historical analyses to visualize and trend analyze and monitor crops over a long period.
4. **Integration and Reporting Module**
This block brings together the results of all the modules together in a single diagnostic report. It also serves as the communication channel of down-stream services like advisory generation or external API integration. The modular framework enables the further development like the real-time alerts or the changes toward the continuous monitoring by the IoT.

B. Design Considerations

The system was developed according to a number of principles:

- **Modularity:** Every part has its own work, and hence a deep learning model, severity algorithm or interface can be updated and without interfering with other parts of the pipeline.
- **Scalability:** The architecture will be able to handle larger datasets, more types of crops and more user loads.
- **Maintainability:** Logical separation enables the easier maintenance of a program and subsequent improvements.
- **User-Centric Output:** The design provides that complex machine learning computations are hidden and given straightforward results to users in the form of a simple interpretation.
- **Security:** The storage of user information, diagnostic history, and uploaded images is done by means of secure data storage and access control.

IV. METHODOLOGY

A. System Workflow Overview

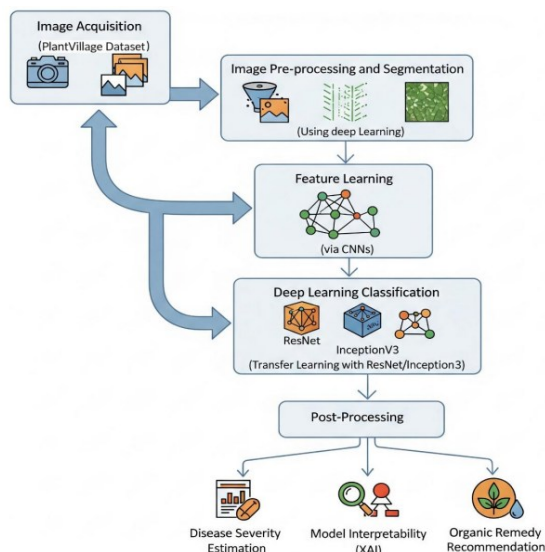


Fig 1. Proposed Deep Learning–Based Plant Disease Diagnosis Architecture.

The system suggested is based on a multi-step working process, which contains data acquisition, pre-processing, features extraction, disease classification, and post-processing to estimate the severity. Each phase will be independent and aimed at providing a part of a diagnostic pipeline. Fig. 1 represents the general architecture that is taken in this work.

The input images were in the form of leaf images that underwent preprocessing to improve the visual images and standardizing the dimensions. These images are then fed to a deep learning classifier which is of ResNet50 architecture. The post-processing phases measure the amount of infection severity and package the output to be interpreted.

B. Dataset Construction and Preparation

The combination of samples of the PlantVillage and DrPlantify resulted in a single set of 49 disease classes of the most important crops, such as tomato, apple, potato, maize, grape, and bell pepper.

Table 1 shows the summary of the dataset after fusion.

Table 1: Dataset Summary After Fusion.

Component Dataset	No. of Classes	Approx. Images	Characteristics
PlantVillage	38	~54,000	Lab-quality, uniform background
DrPlantify	11	~8,000	Field-quality, variable lighting
Combined Dataset	49	~62,000	Mixed environment, higher variance

The fusion of data added variations like shadows, inconsistent angles and noise to the system, enhancing the resistance of the system to the real-world input.

Cleaning and Preprocessing of Data.

- Elimination of redundant and low quality samples.
- Standard resizing to 224×224
- ImageNet image normalization in ImageNet mean and standard deviation.
- A stratified 5-fold cross-validation strategy was employed to ensure robust evaluation and reduce sampling bias across disease categories.

Techniques of Data Augmentation.

The augmentations used in order to prevent overfitting and enhance better generalization are as follows (Table.2):

Table 2: Techniques of augmentation.

Augmentation Type	Purpose
Random Rotation	Simulates leaf variance.
Horizontal/Vertical Flip	Spatial invariance is strengthened.
Color Jitter	Manages the discrepancies in lighting.

Brightness/Contrast Shift	Shift Simulates the outside world.
Random Zoom / Crop	Simulates variations of distance.

C. Image Pre-processing and Segmentation

All the images are taken through a preprocessing pipeline before being classified which makes them more suitable to deep learning analysis. Segmentation can be used in isolation of the leaf structures against cluttered backgrounds.

Steps include:

1. RGB → HSV conversion
2. Gaussian blur Noise filtering.
3. Green leaf masking Threshold-based green leaf masking.
4. Computation of lesion area severity by isolating lesion areas.

These measures will take care of the fact that the classifier gets regular and noise-free input images.

D. Deep Learning Classification Model

A transfer learning strategy was followed with ResNet50 that is trained on ImageNet. Remnant connections enable effective training and still maintain representation of the spatial features.

The last layers were substituted with: to fit the model to 49 categories of plant-diseases (Table.3).

- Fully Connected Layer (2048 → 512)
- ReLU activation
- Dropout (0.3)
- Fully Connected Layer (512 → 49)

Table 3: Training Configuration

Parameter	Value
Optimizer	Adam
Learning Rate	0.001 (decay on plateau)
Batch Size	32
Epochs	20
Loss Function	Cross-Entropy
Validation Split	20%

The maximum validation accuracy was 81.07% obtained. The minimal decrease in accuracy between PlantVillage-only training and dataset fusion training is evidence of the added realism and complexity of dataset fusion, which leads to improved field performance[15].

E. Severity Estimation Pipeline

The HSP color space is useful in isolating diseased areas because chromatic content (H) and illumination (V) are separated (Table.4).

Severity Calculation Formula

$$\text{Severity(\%)} = \left(\frac{\text{Total Leaf Pixels} - \text{Healthy Green Pixels}}{\text{Total Leaf Pixels}} \right) \times 100$$

Table 4: Severity Levels

Severity Range	Stage
0–10%	Healthy
10–25%	Mild

25–50%	Moderate
>50%	Critical

F. Post-Processing and Interpretation Layer

The disease label and the severity score are directed to the interpretation module which translates the system output to:

- Biological explanation
- Visual symptom checklist
- Suggestions of organic treatment.
- Preventive measures

This element aids in making decisions and does not change the classification algorithm.

V. RESULTS AND ANALYSIS

The proposed system was tested on the basis of various dimensions, such as the accuracy of model classification, reliability of the severity estimates and end-to-end user experience by a deployed web interface. The part includes the quantitative and qualitative results of the developed solution.

A. Model Performance Evaluation.

The ResNet50 classifier trained on the fused PlantVillage + DrPlantify dataset was able to converge on a fixed point and had high generalization on 49 disease classes. The last validation accuracy of 81.07 was obtained and it is indicative of the added complexity and real world variation using dataset fusion (Table.5 and 6).

Table 5. Final Validation Accuracy.

Metric	Value
Validation Accuracy	81.07%
Validation Loss	0.48
No. of Classes	49
Training Epochs	20

Table 6. Class-wise Performance Metrics (Representative Classes).

Class Name	Precision	Recall	F1-Score	Support
Apple Scab	0.83	0.79	0.81	1020
Tomato Late Blight	0.85	0.82	0.83	980
Corn Rust	0.80	0.76	0.78	740
Grape Black Rot	0.84	0.81	0.82	860
Healthy Leaves	0.88	0.90	0.89	2100

The confusion matrix further confirms that most misclassifications occur among visually similar diseases, particularly under imbalanced class distributions.

Compared to field-captured and lab-captured images, this accuracy is competitive because it is difficult to deal with domain shift in this case. The confusion matrix demonstrated that the model had better performance with regard to separating fungal and bacterial patterns, but there were some visually similar diseases with marginal confusion.

B. System Interface and User Experience.

The system developed has a fully realized web interface allowing the smooth interaction of the user. Fig. 2 indicates the landing page that directs users to start scanning leaves of plants in order to be diagnosed.

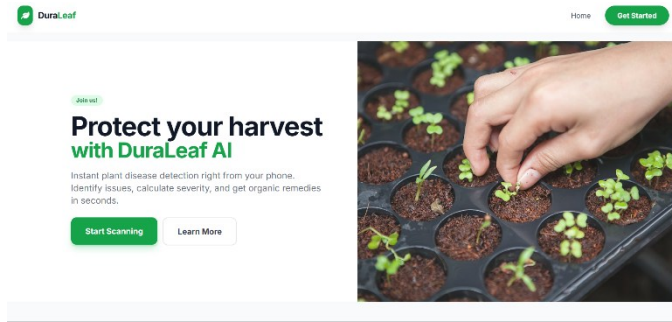


Fig 2. User Interaction and Landing Page of the System.

The design is focused on simplicity and clarity so that one can start scanning on the homepage.

C. Authentication of the user and managing his profile.

Fig. 3 shows the user registration interface that allows the easy and secure onboarding and personalized scan history.

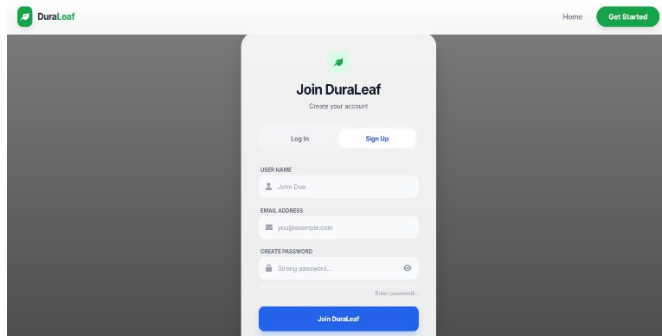


Fig 3. Registration and Sign-Up Interface of the user.

This module will guarantee that the results of the diagnostic are user-related, allowing the monitoring of the health of crops over a long period of time.

D. Class-wise Performance and Imbalance Analysis

To provide a deeper insight into the classification performance, class-wise precision, recall, and F1-score were computed for all 49 plant-disease categories. The dataset exhibits moderate class imbalance, particularly for diseases originating from the DrPlantify dataset, which contain fewer samples compared to PlantVillage classes. This imbalance primarily affected recall values for minority classes. To mitigate this issue, stratified sampling and data augmentation techniques were employed during training, improving minority-class sensitivity.

E. Leaf Image Upload and Scan Processing.

The authentication sends users to the scanning interface (Fig. 4) in which images may be uploaded to be analyzed.

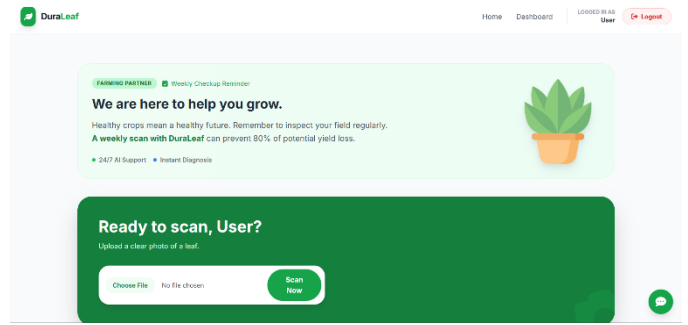


Fig 4. Uploading panel to start disease detection by use of leaf image.

The interface transfers the image to the backend inference pipeline instantly, resulting in preprocessing, classification and estimation of the severity.

E. Diagnostic History and Record Tracking.

The system keeps a record of every scan by the user. The history panel is provided in Fig. 5 and displays the past analyses.

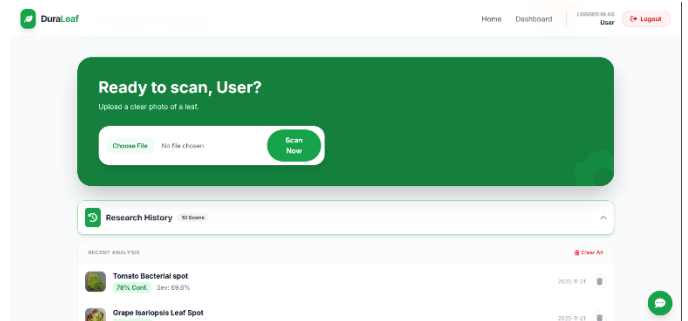


Fig 5. Historical Diagnosis with Confidence Scores and Severity Percentages.

The feature will enable farmers to monitor conditions of recurring diseases, guess the spread patterns, and analyze variations in seasons.

F. A.I.-Generated Diagnostic Report.

The entire diagnostic report prepared by the system is presented in Fig. 6. This is the fundamental product of the model and it is a combination of:

- Predicted disease
- Confidence score
- Percentage of severity of infection.
- Symptoms summary
- Recommendations of organic remedies.

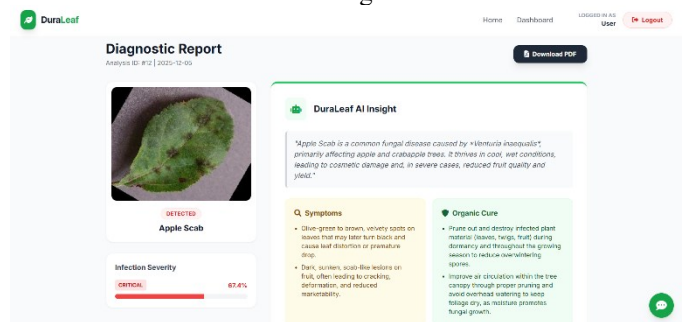


Fig 6. Produced Diagnostic Report and its Severity Assessment and Treatment Recommendation.

The report has some practical recommendations that can be directly applied by farmers without the involvement of experts.

G. Chatbot Support and Advisory Service.

The system has a conversational assistant, besides automated diagnosis. Fig. 7 represents a sample query-answer interaction.

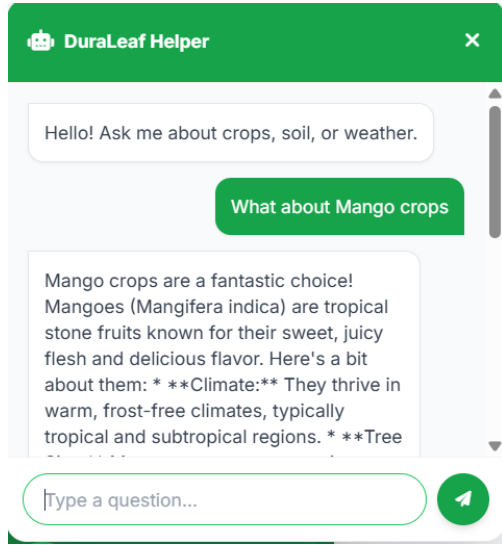


Fig 7. Smart Agricultural Chatbot that Delivers Farm Strategy advice to the user queries.

The chatbot is context-sensitive, and improves the perception of the user by explaining crops, soil management, symptom of the disease, and the environmental conditions.

The 5-fold cross-validation yielded an average accuracy of 80.62% with a standard deviation of $\pm 1.14\%$, indicating consistent generalization across different data splits (Table.7).

Table 7: Robustness Evaluation Under Environmental Variations

Condition	Accuracy (%)
Normal lighting	81.07
Low illumination	78.92
High brightness	79.40
Cluttered background	77.85

H. Hybridization of Severity Estimation Classification Product.

The system combines the quantitative approximation of severity with the classification of diseases in order to produce more information. This particularly comes in handy when urgency of treatment is to be determined (Table.8).

Table 8: Findings of Severity Levels.

Severity (%)	Condition	Recommended Action
0–10%	Healthy	Routine monitoring
10–25%	Mild	Apply preventive spray
25–50%	Moderate	Start specific treatment.
>50%	Critical	Intervention needs to be done immediately.

The seriousness of Fig. 6 (e.g., 67.4%) points at those instances, in which some immediate remediation is required.

VI. CONCLUSION

The paper has provided an in-depth deep learning-based plant disease detection and severity estimation framework, to facilitate precision agriculture by providing scalable and affordable diagnostic tools. The system combined the PlantVillage and DrPlantify data and could identify 49 categories of plants-disease and become more resilient to real-life differences. The optimized ResNet50 model showed high classification of results and the severity estimation module using HSV gave an objective, quantitative measure of the infected leaf area. Moreover, the incorporation of an AI-assisted online interface and historical tracking, alongside an online interface, makes the product more usable by providing advice to end users in an actionable and easy to read manner. Contrary to the traditional methods where only the disease is classified, the proposed platform is a holistic decision-support framework because it integrates the disease identification, quantification of severity, and treatment advice into one platform. The experimental findings prove that the system can help farmers to monitor the health of crops, choose between interventions, and minimize losses in yields due to the delay or wrong diagnosis. The modular design also provides a modular architecture that is easy to integrate with the new agricultural technologies, and this design is also scalable and maintainable. An upcoming research will be aimed at broadening the system to encompass a more comprehensive identify varieties of crops and types of diseases and new datasets to capture larger-scale crops images by farmers to further expand realistic generalization. Early warning and predictive capability can be enhanced by integration with real-time field data sources like IoT sensors and weather information. Moreover, the implementation of lightweight and edge-optimized models will allow to provide offline inference on mobile and low-power hardware and make the framework more accessible in remote and resource-constrained agricultural settings.

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